IMAGE PROCESSING AND ITS APPLICATION

MODULE - 1

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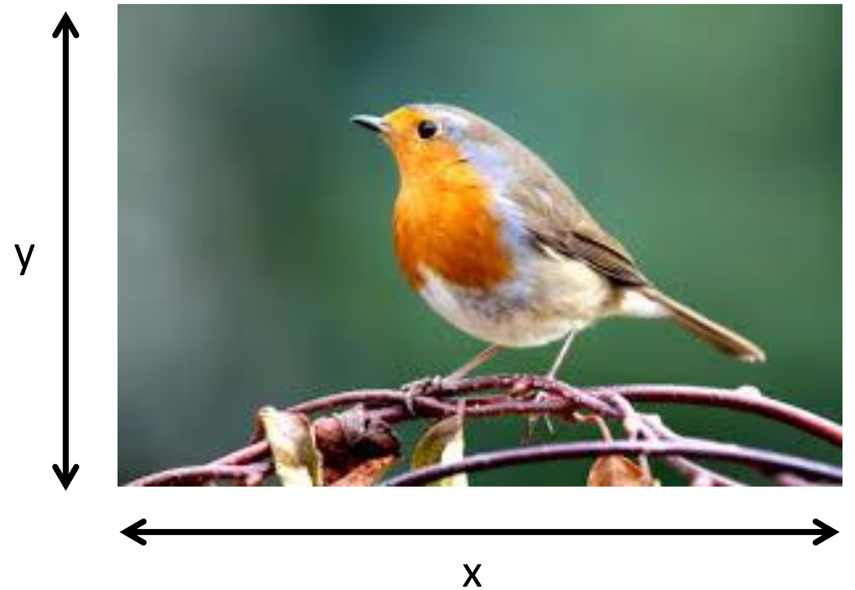
Image

A **digital image** is a two-dimensional function



X = width
Y = height
(x,y) = width,height

Image

- A **digital image** is a two-dimensional function

  $$f(x,y)$$

  x, y → *spatial coordinates*

  f(x,y) → *intensity / gray level at the point*

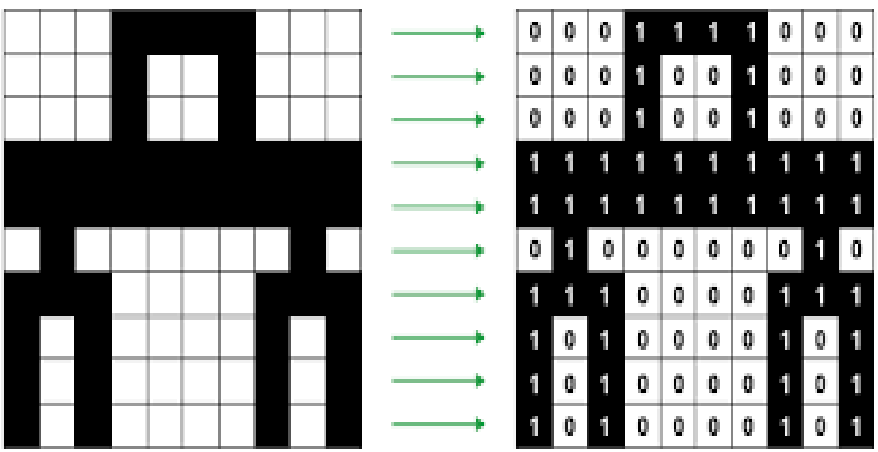
- An image is made up of **pixels (picture elements)**.

Digital Image Processing

- Digital image processing is the manipulation of digital images by mean of a computer.
- It has two main applications:
  - Improvement of pictorial information for human interpretation
  - Processing of image data for storage, transmission and representation for autonomous machine perception.

Types of Images

- **Binary Image** – Only two values (0 & 1)
- **Grayscale Image** – Shades from black to white (0–255)

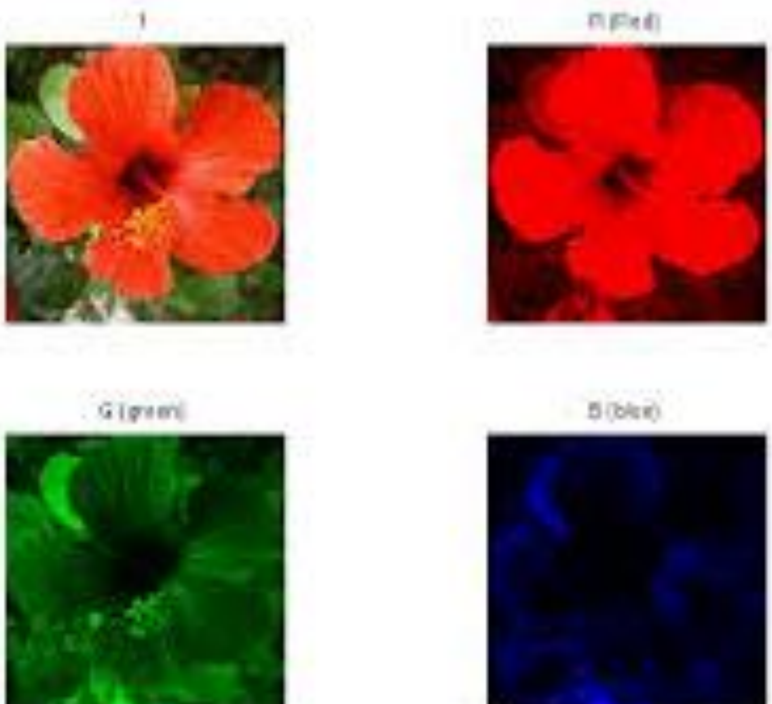


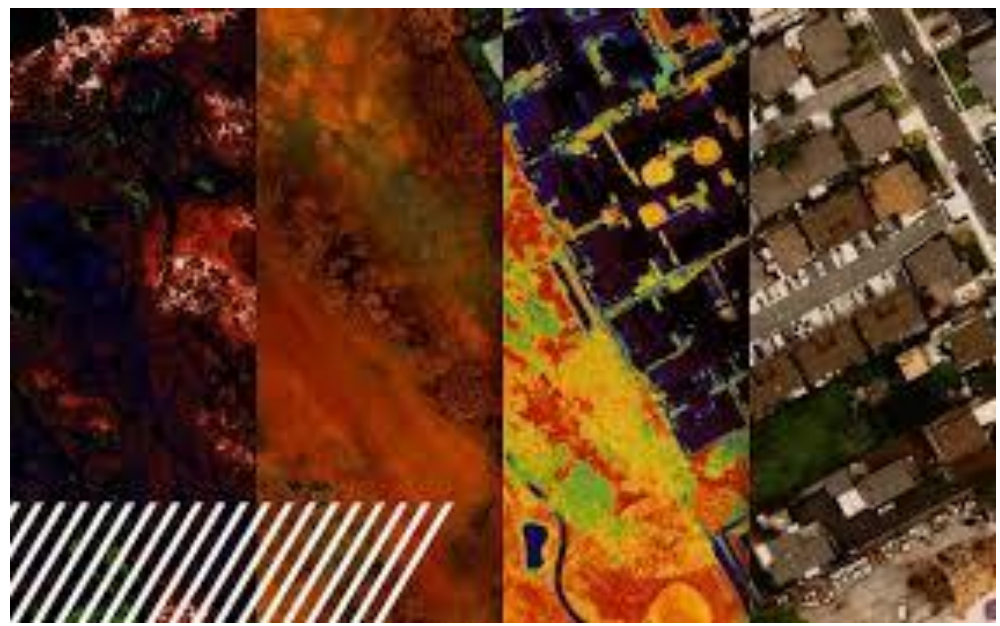
Binary Image

Greyscale Image

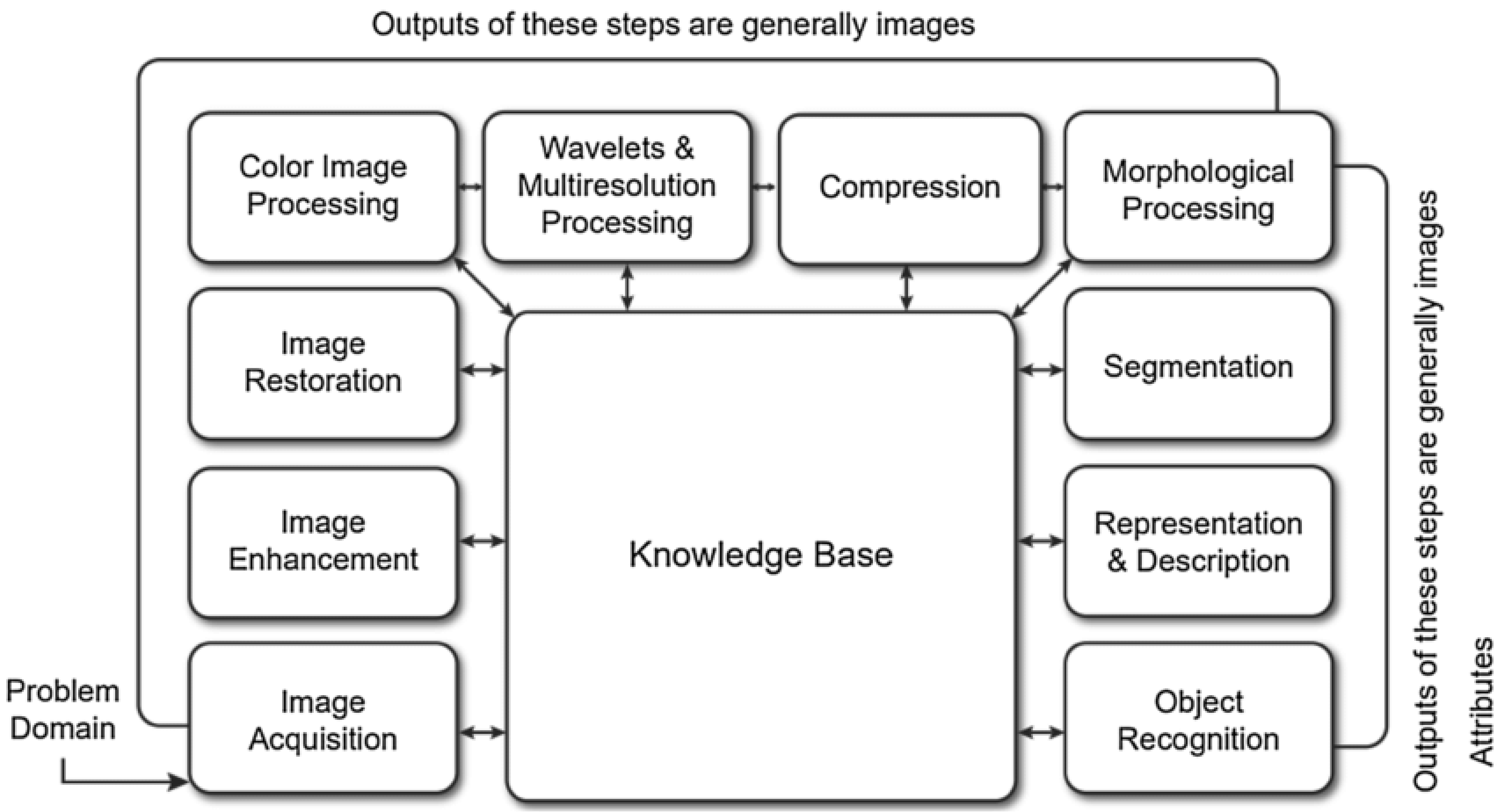
Types of Images

- **Color Image** – Combination of RGB components
- **Multispectral Image** – More than 3 bands (satellite images)



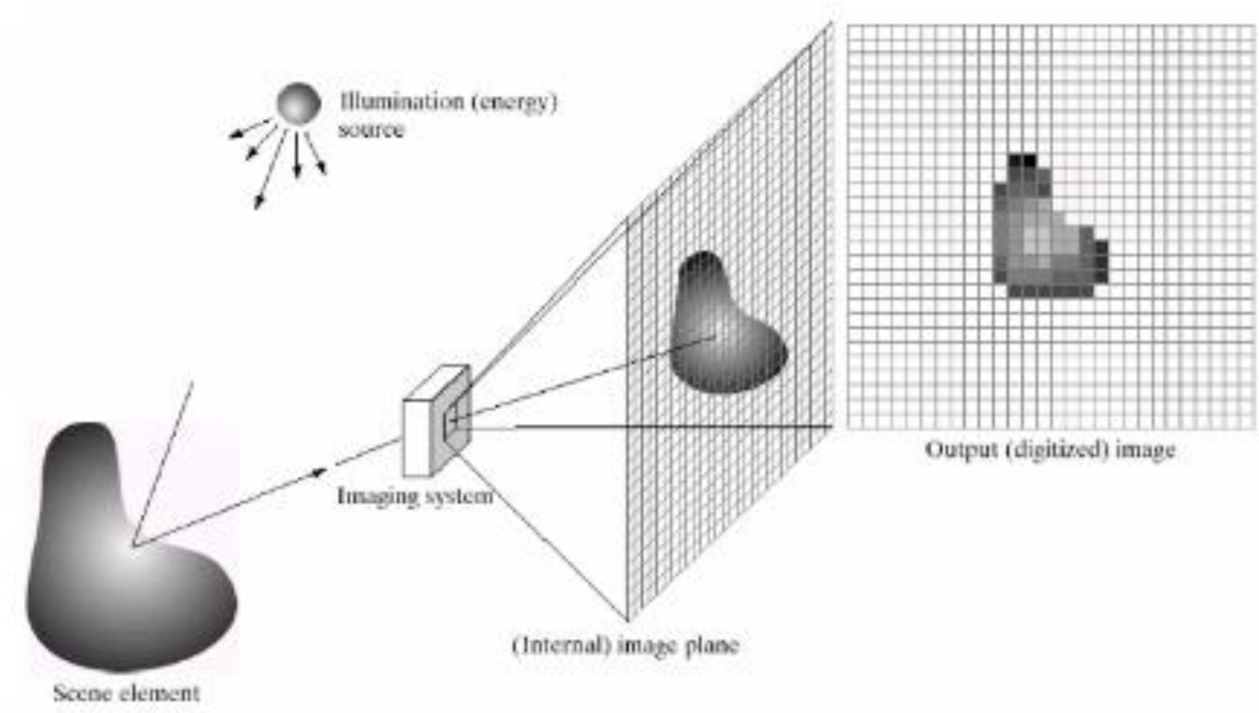


Fundamental steps in image processing



1. Image Acquisition

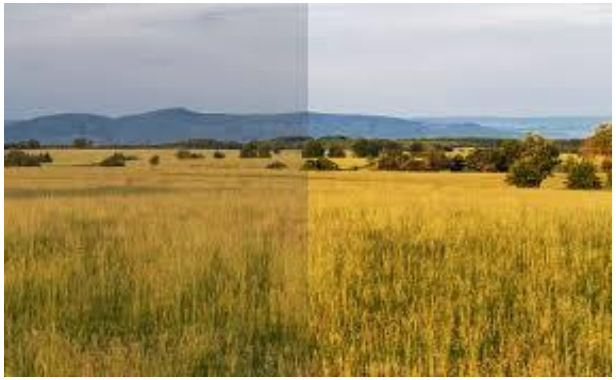
This involves capturing an image using a digital camera or scanner, or importing an existing image into a computer.



2. Image enhancement

- Improving the visual quality of an image, such as increasing contrast, reducing noise, and removing artifacts.
- **Techniques:**
  – Contrast stretching
  – Histogram equalization
  – Image smoothing
  – Image sharpening

- **Applications:**
  – Improving brightness in medical images
  – Enhancing satellite images

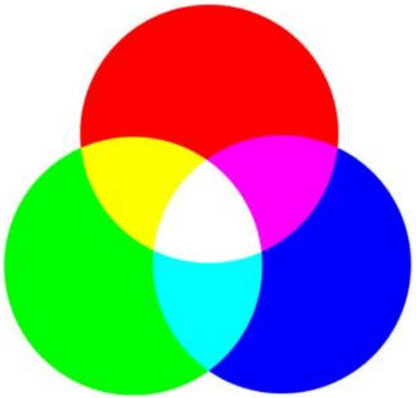


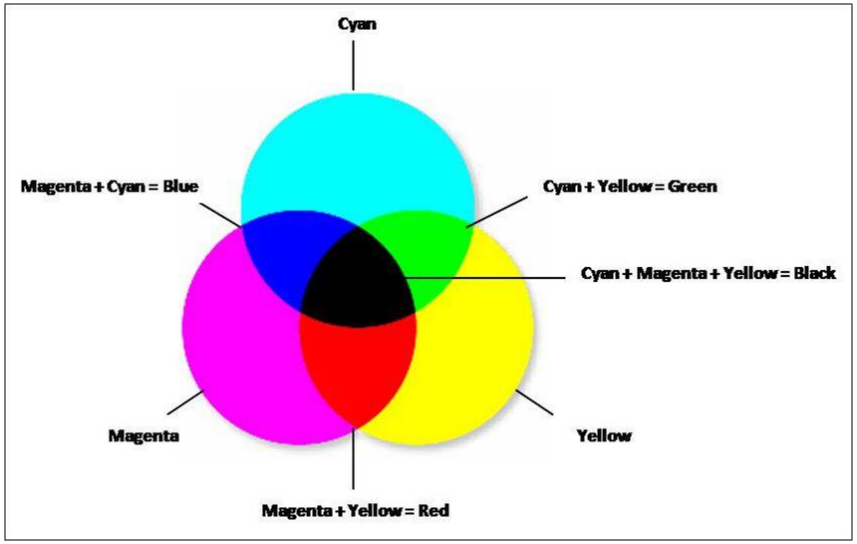
3. Image restoration

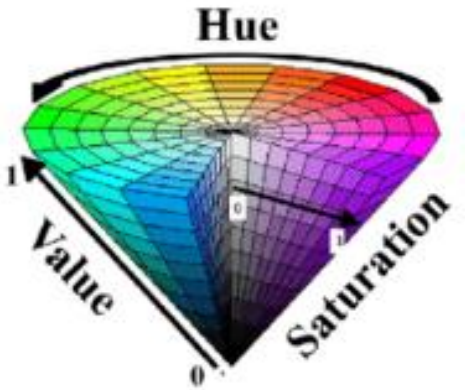
- Removing degradation from an image, such as blurring, noise, and distortion.
- **Common Degradations:**
  - Noise
  - Motion blur
  - Defocusing
- **Techniques:**
  - Wiener filtering
  - Inverse filtering

Color Image Processing

- This step deals with processing color information in images.
- Color models like **RGB, HSV, CMY** are used.
- Helps in better interpretation and feature extraction.
- **Hue** is the color component, representing different colors based on angle ranges within the HSV cone

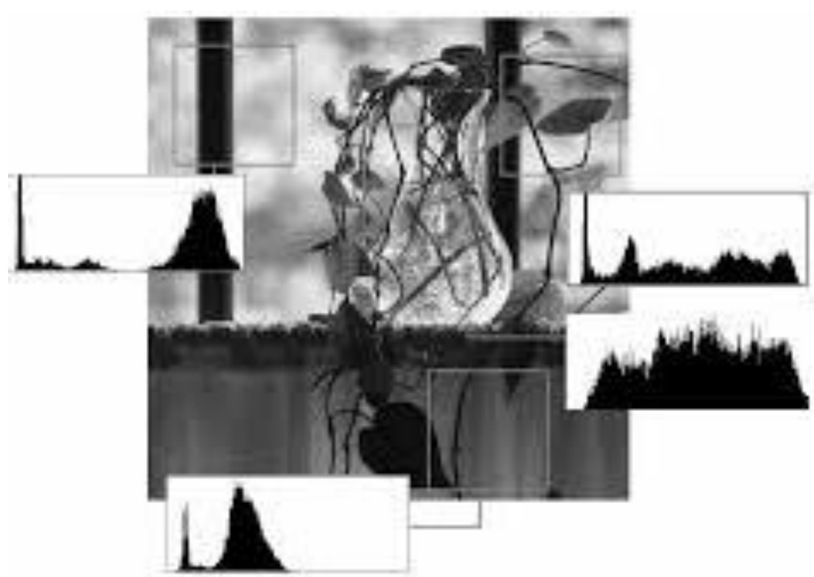






5. Wavelets and Multiresolution Processing

- Wavelet transform represents images at multiple resolutions.
- Useful for compression and feature extraction.
- **Applications:**
  - Image compression (JPEG2000)
  - Texture analysis



6. Image Compression

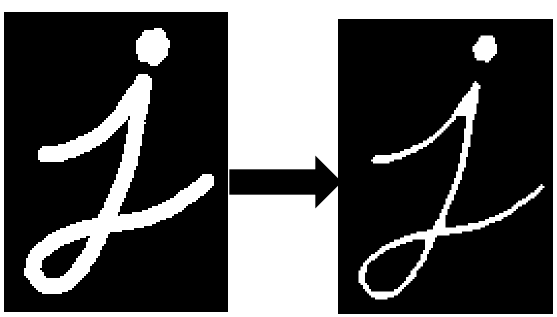
- Image compression reduces the amount of data required to store or transmit an image.
- **Explanation:**
  - Removes redundancy in image data.
  - Important for storage and communication.
- **Types:**
  - **Lossless compression** – No loss of data (Huffman coding)
  - **Lossy compression** – Some data loss (JPEG)

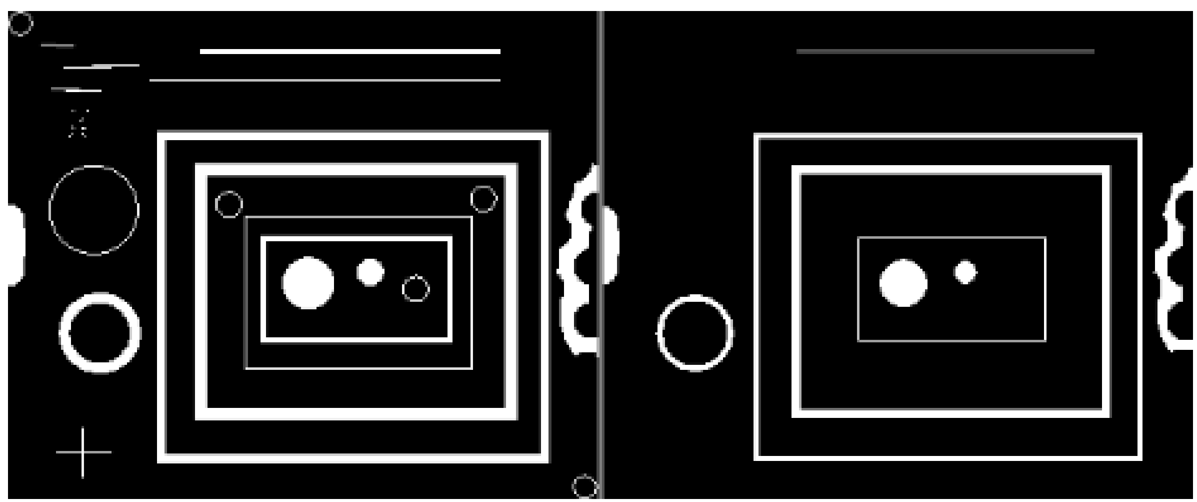
7. Morphological Processing

- Morphological processing focuses on the shape and structure of objects in an image. Mainly used for binary images.
- **Operations:**
  - Erosion
  - Dilation
  - Opening
  - Closing
- **Applications:**
  - Noise removal
  - Boundary detection

7. Morphological Processing - Erosion

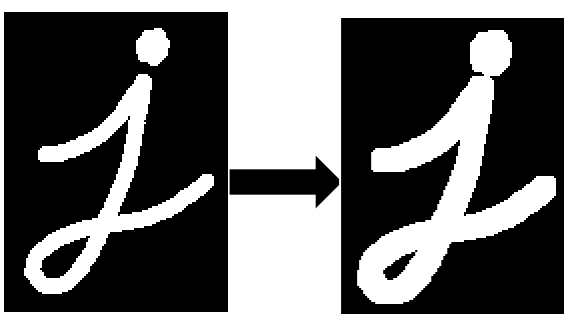
- Erosion is a fundamental morphological operation that reduces the size of objects in a binary image. It works by removing pixels from the boundaries of objects.
- **Purpose:** To remove small noise, detach connected objects, and erode boundaries.

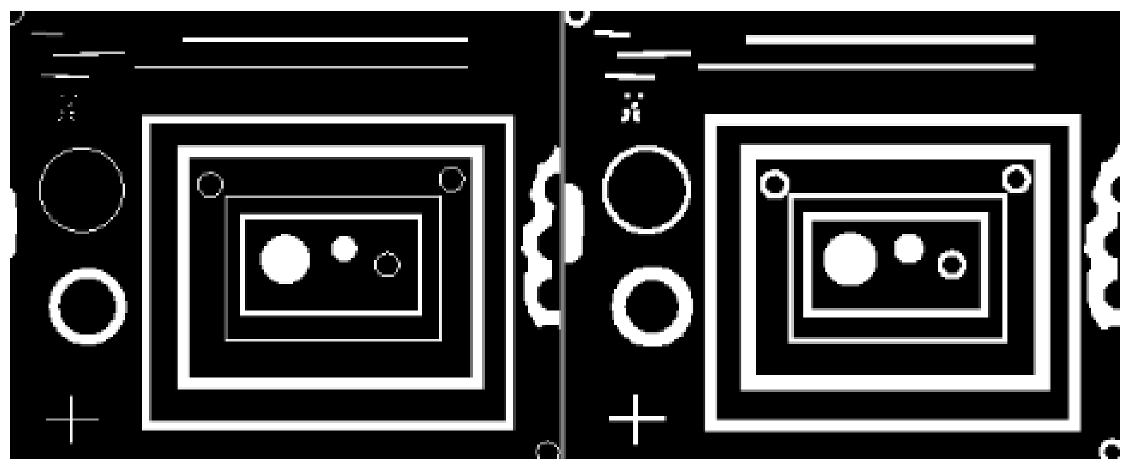




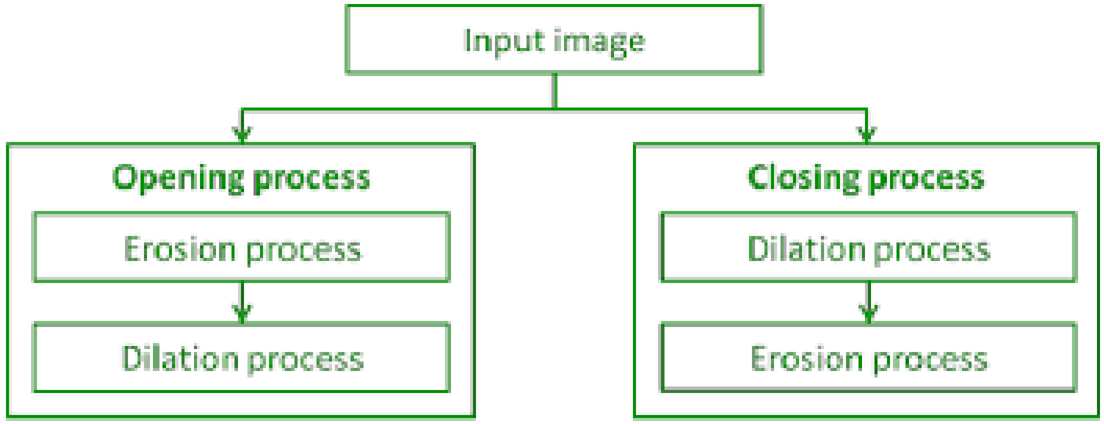
7. Morphological Processing - Dilation

- Dilation is the opposite of erosion and is used to increase the size of objects in an image.
- **Purpose:** To join adjacent objects, fill small holes, and enhance features.



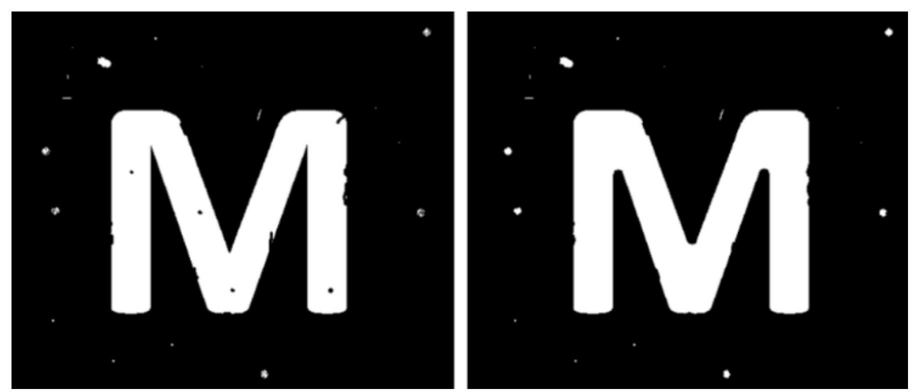


7. Morphological Processing



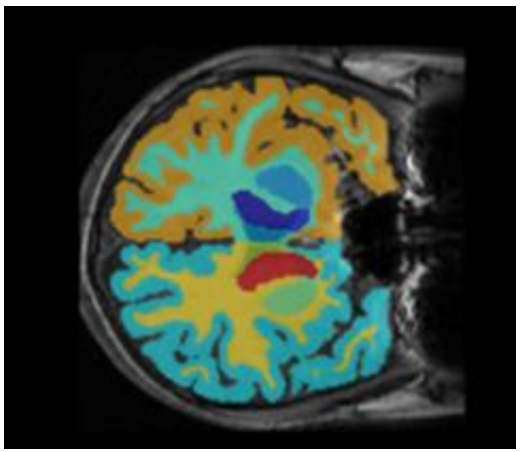
To remove small objects or noise from the image while preserving the shape and size of larger objects.

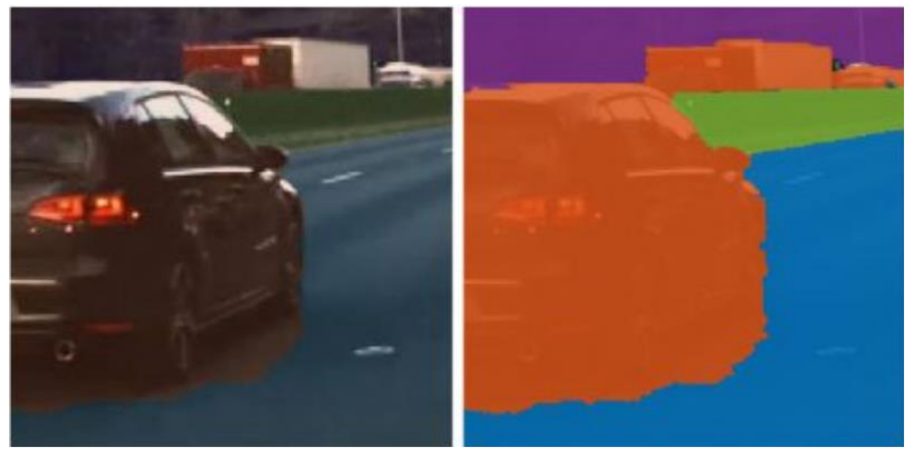
To fill small holes and gaps in objects while preserving their overall shape.

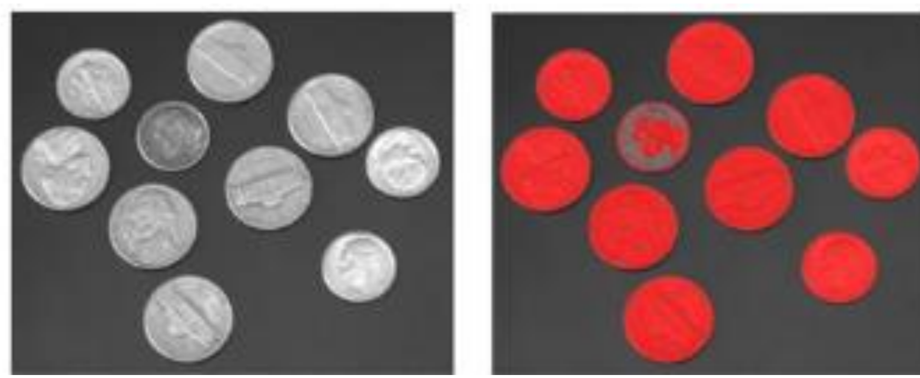


8. Image Segmentation

- Image segmentation involve separating foreground from background or clustering regions of pixels based on similarities in color or shape.

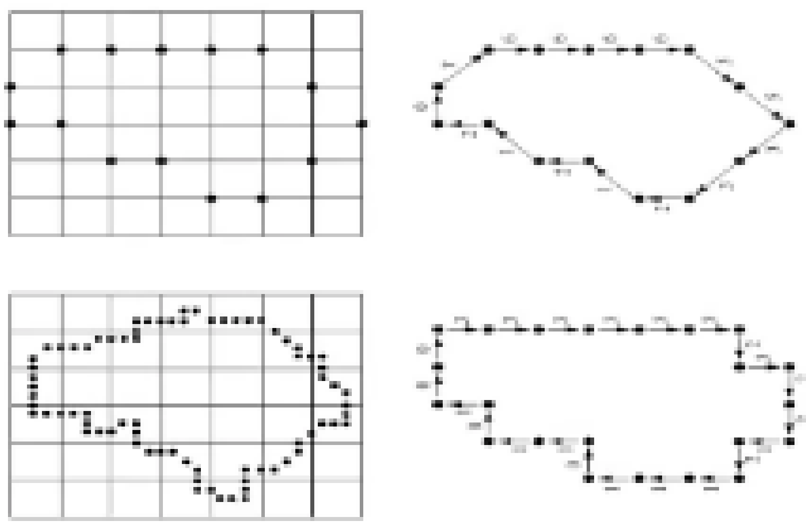




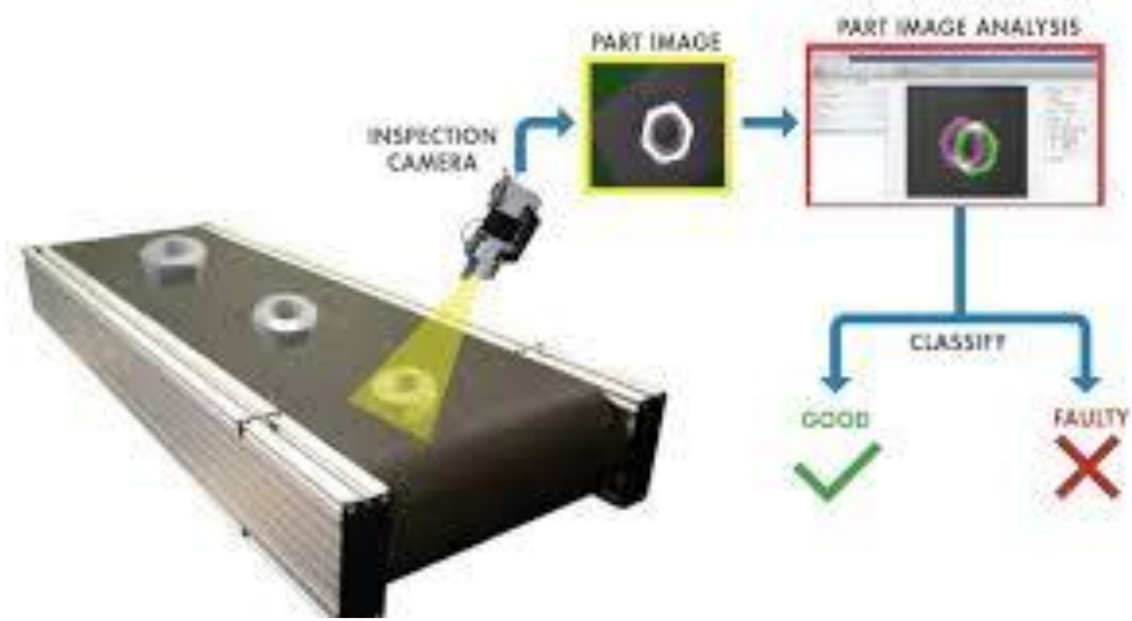


9. Representation and Description

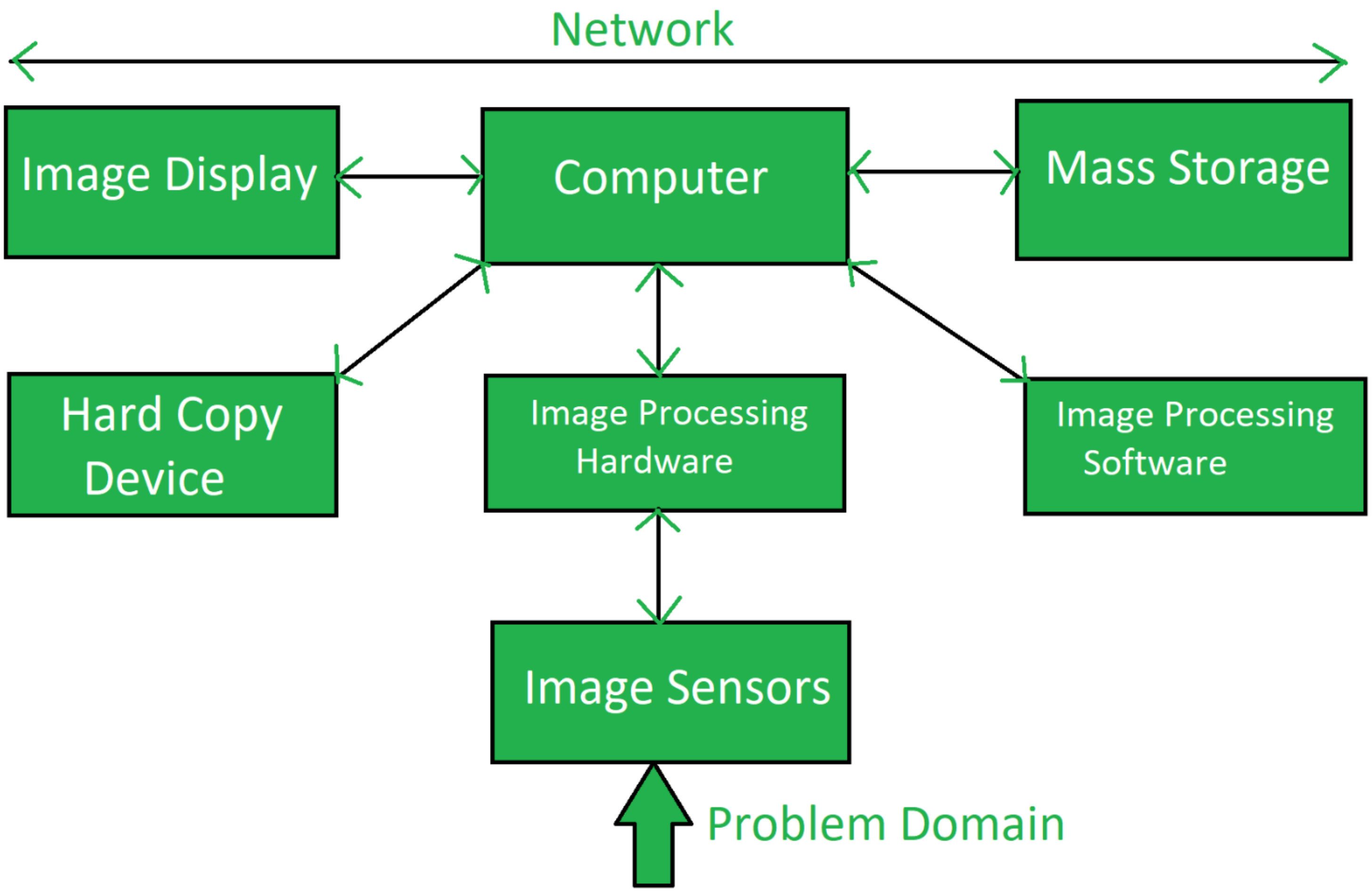
- Raw data consists of boundaries of a region or all the points in the region
- Boundary Representation → focus on external shape
- Region Representation → focus on internal characters such as texture



10. Recognition and Interpretation



Components of Image Processing system



Components of Image Processing system

- Image Sensor
  - Captures the image from the physical world.
  - Converts optical signals (light) into electrical signals.
  - Consists of a **sensor** and **signal conditioning unit**.

Components of Image Processing system

- **Image Processing Hardware:** Image processing hardware is the dedicated hardware that is used to process the instructions obtained from the image sensors.
- **Image Processing Software:** Image processing software is the software that includes all the mechanisms and algorithms that are used in image processing system.

Components of Image Processing system

- **Computer:** Computer used in the image processing system is the general purpose computer that is used by us in our daily life.
- **Mass Storage:** Mass storage stores the pixels of the images during the processing.
- **Hard Copy Device:** Once the image is processed then it is stored in the hard copy device. It can be a pen drive or any external ROM device.
- **Image Display:** It includes the monitor or display screen that displays the processed images.
- **Network:** Network is the connection of all the above elements of the image processing system.

Color Model

1. RGB
2. CMYK
3. HSV
4. YIQ

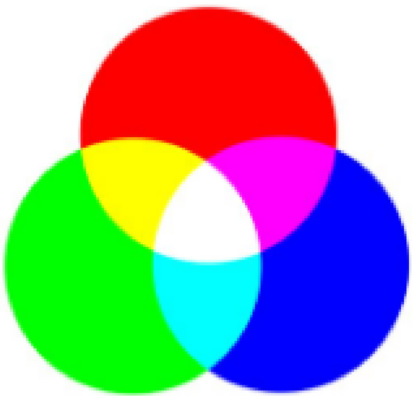
Color Model - RGB

- The RGB color model is the most prevalent model used in digital image processing and OpenCV.
- A color image in this model consists of three channels: one for each primary color—Red, Green, and Blue.
- All other colors are created through varying proportions of these three colors.
- A value of 0 represents black, and as the value increases, the color intensity increases.

Color Model - RGB

Properties:

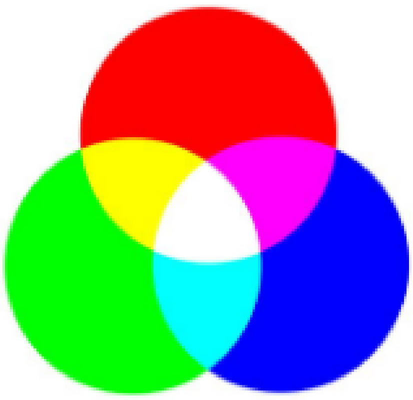
- It is an additive color model, where colors are added to black.
- It has three main channels: Red, Green, and Blue.
- It is used in digital image processing, OpenCV, and online logos.



Color Model - RGB

Colour combination:

- Green(255) + Red(255) = Yellow
- Green(255) + Blue(255) = Cyan
- Red(255) + Blue(255) = Magenta
- Red(255) + Green(255) + Blue(255) = White

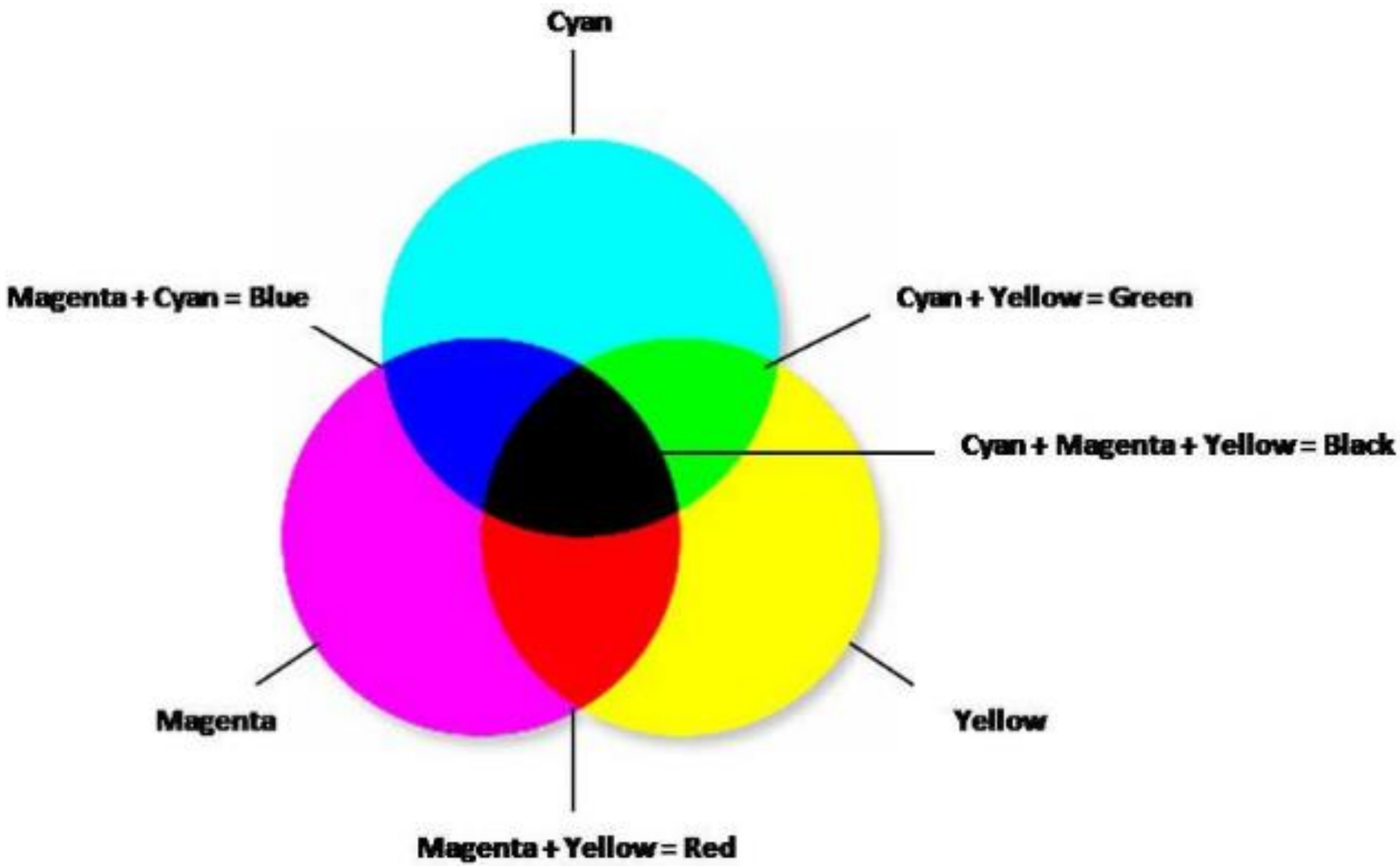


Color Model - CMYK

- The CMYK color model is extensively used in the printing industry.
- The acronym CMYK stands for Cyan, Magenta, Yellow, and Key (Black).
- Unlike the RGB model, which is additive, the CMYK model is subtractive.
- This means it works by subtracting varying degrees of light absorbed by the colors.
- 0 represents the primary color at its full intensity.
- 1 represents the absence of color, resulting in white.

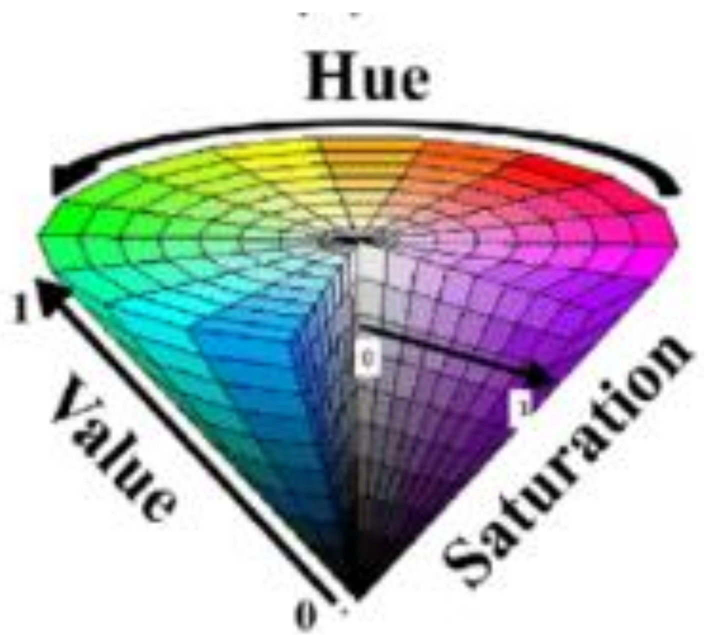
Color Model - CMYK

- Cyan is the negative of Red.
- Magenta is the negative of Green.
- Yellow is the negative of Blue.



HSV Color Model

- The HSV color model represents colors through three channels: Hue, Saturation, and Value.
- HSV aligns with how humans perceive color.
- The HSV model is often depicted as a cone.



HSV Color Model

- **Hue** is the color component, representing different colors based on ange ranges within the HSV cone:
  - Red falls between 0 and 60 degrees.
  - Yellow falls between 61 and 120 degrees.
  - Green falls between 121 and 180 degrees.
  - Cyan falls between 181 and 240 degrees.
  - Blue falls between 241 and 300 degrees.
  - Magenta falls between 301 and 360 degrees.

HSV Color Model

- **Saturation** describes the purity of the color, ranging from 0 to 1.
- A saturation of 0 results in a shade of gray, while a saturation of 1 corresponds to a fully saturated color.
- This parameter helps determine the amount of gray in the color.

HSV Color Model

- **Value** indicates the brightness or intensity of the color, ranging from 0% to 100%.
- A value of 0% represents black, while a value of 100% represents the brightest form of the color.
- The higher the value, the more vivid and intense the color appears.

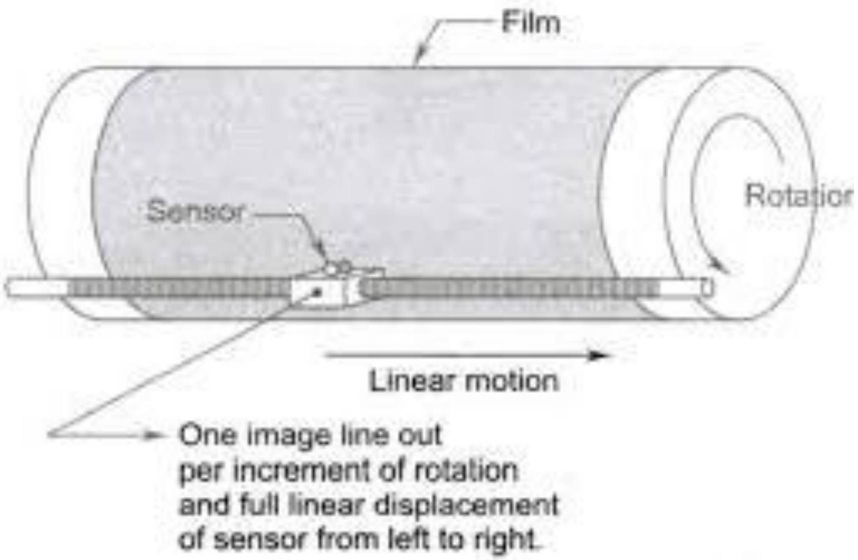
Image sensing and acquisition

- Image acquisition is the first step in digital image processing.
- To acquire a digital image, two elements are required.
  - The first is a physical device that is sensitive to a band in the electromagnetic energy spectrum, and that produces an electrical signal output proportional to the level of energy sensed.
  - The second called digitizer is a device for converting electrical output of the physical sensing device into digital form.

Image sensing and acquisition

1. Image acquisition using a single sensor
2. Image acquisition using sensor strip
3. Image acquisition using sensor arrays

Image acquisition using a single sensor



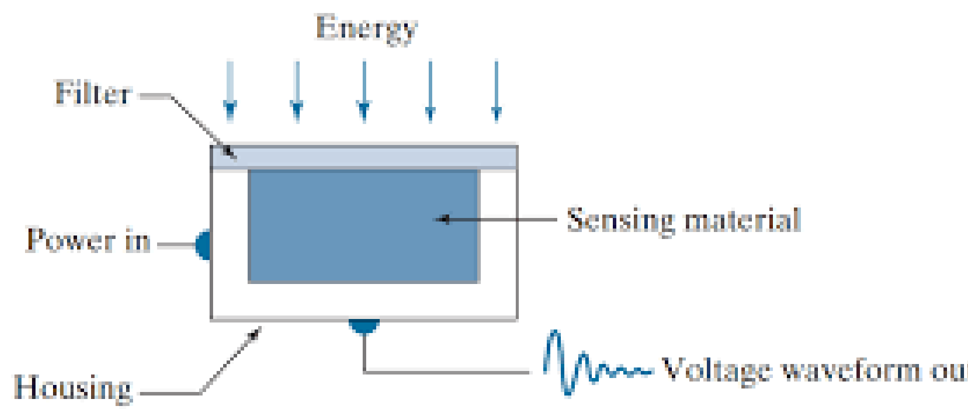


Image acquisition using a single sensor

- Image sensor used in digital cameras is a devise that converts an optical image to an electric signal.
- An image sensor is typically a charge couple devise or complementary metal oxide semiconductor(CMOS) sensor
- Both types can give very good results
- On the surface of these fingernail-size silicon chips are millions of photosensitive diodes ,called photosites, each of which captures a single pixel in the photograph to be.

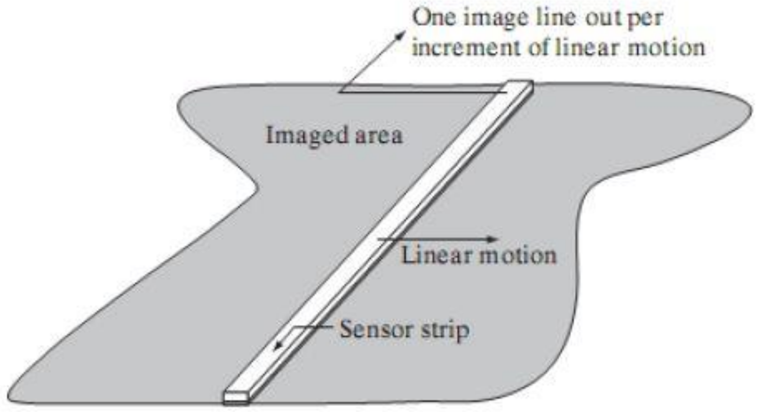
The use of a filter in front of a sensor improves selectivity.

- For example, a green (pass) filter in front of a light sensor favors light in the green band of the color spectrum. As a consequence, the sensor output will be stronger for green light than for other components in the visible spectrum.

Image acquisition using a single sensor

- In order to generate a 2-D image using a single sensor, there have to be relative displacements in both the x- and y-directions between the sensor and the area to be imaged
- The single sensor is mounted on a lead screw, that provides motion in the perpendicular direction. Since mechanical motion can be controlled with high precision, this method is an inexpensive (but slow) way to obtain high-resolution images.
- Other similar mechanical arrangements use a flat bed, with the sensor moving in two linear directions. These types of mechanical digitizers sometimes are referred to as Micro densitometers.

Image acquisition using sensor strip



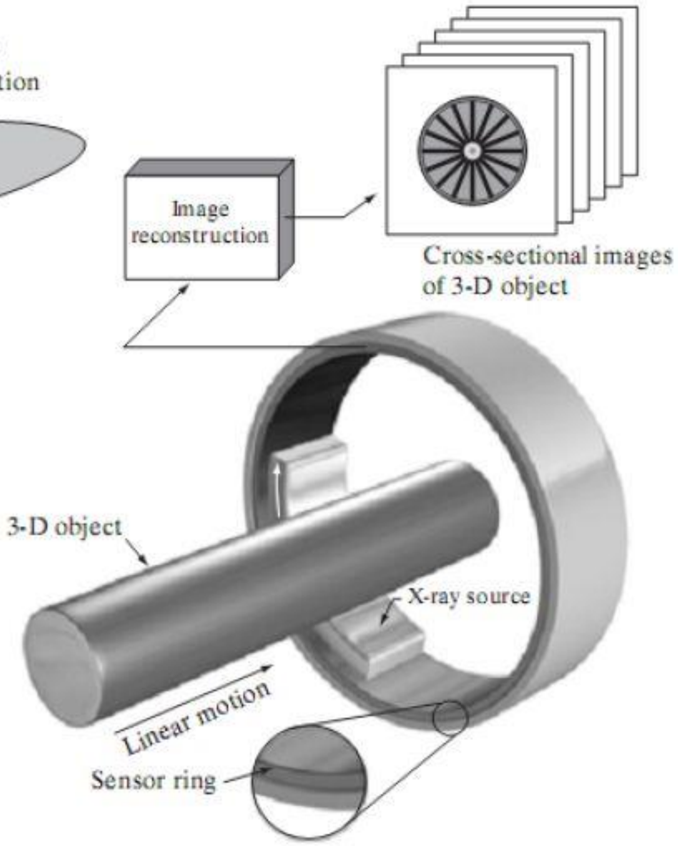


Image acquisition using sensor strip

- The strip provides imaging elements in one direction.
- Motion perpendicular to the strip provides imaging in the other direction
- Sensing devices with 4000 or more in-line sensors are possible.
- One-dimensional imaging sensor strips that respond to various bands of the electromagnetic spectrum are mounted perpendicular to the direction of flight.
- The imaging strip gives one line of an image at a time, and the motion of the strip completes the other dimension of a two-dimensional image.

Image acquisition using sensor strip

- Sensor strips mounted in a ring configuration are used in medical and industrial imaging to obtain crosssectional (“slice”) images of 3-D objects.
- A rotating X-ray source provides illumination and the portion of the sensors opposite the source collect the X-ray energy that pass through the object

Image Acquisition using Sensor Arrays

- This type of arrangement is found in digital cameras.
- A typical sensor for these cameras is a CCD (charged coupled device) array, which can be manufactured with a broad range of sensing properties and can be packaged in rugged arrays of 4000 * 4000 elements or more.
- CCD sensors are used widely in digital cameras and other light sensing instruments.
- The response of each sensor is proportional to the integral of the light energy projected onto the surface of the sensor, a property that is used in astronomical and other applications requiring low noise images.

Image Acquisition using Sensor Arrays

- The first function performed by the imaging system is to collect the incoming energy and focus it onto an image plane.
- If the illumination is light, the front end of the imaging system is a lens, which projects the viewed scene onto the lens focal plane.
- The sensor array, which is coincident with the focal plane, produces outputs proportional to the integral of the light received at each sensor

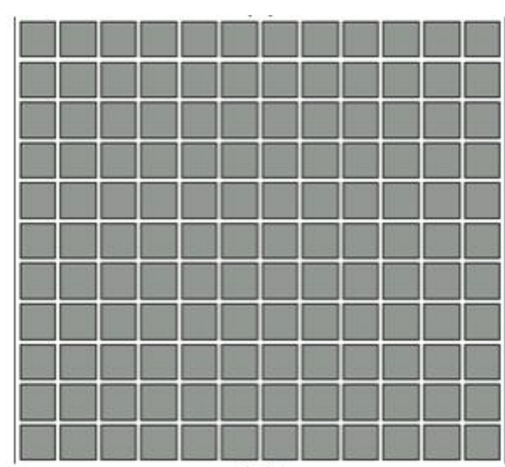


Image Acquisition using Sensor Arrays

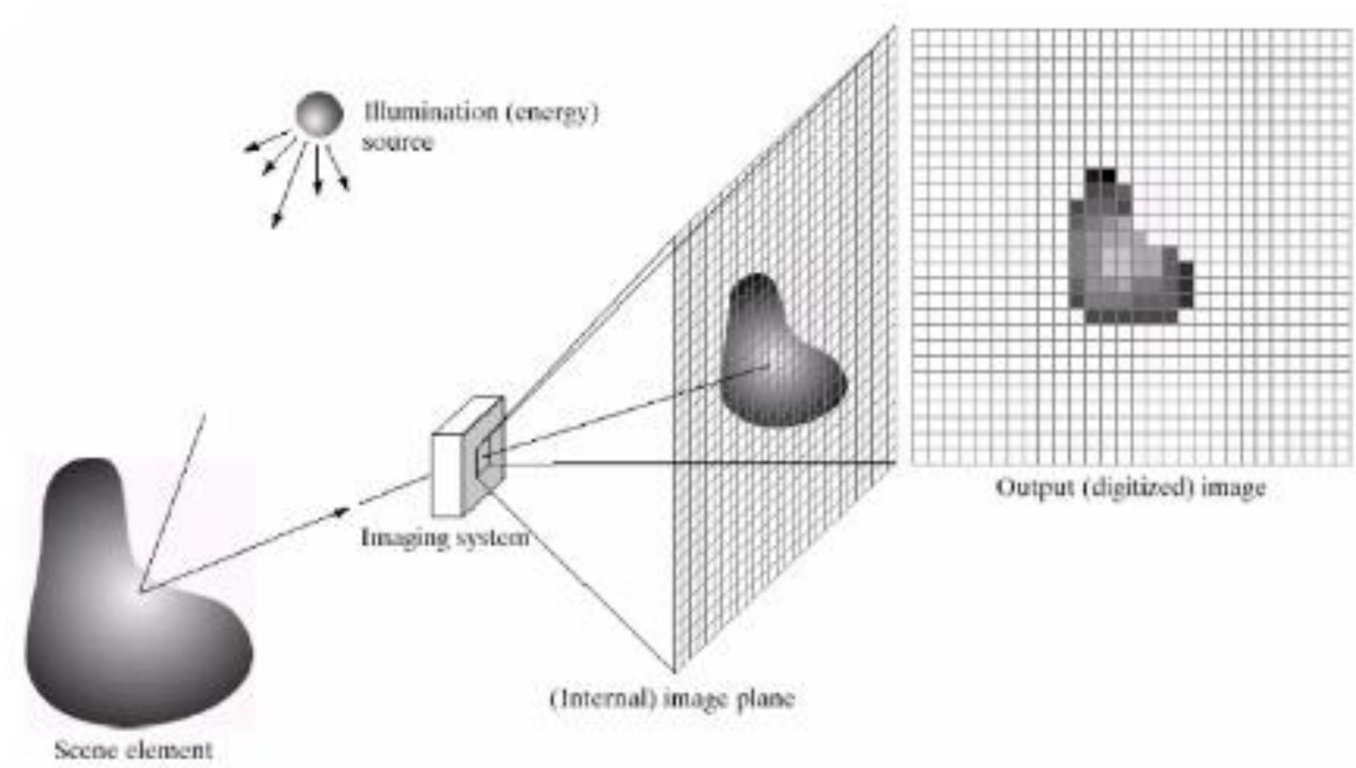


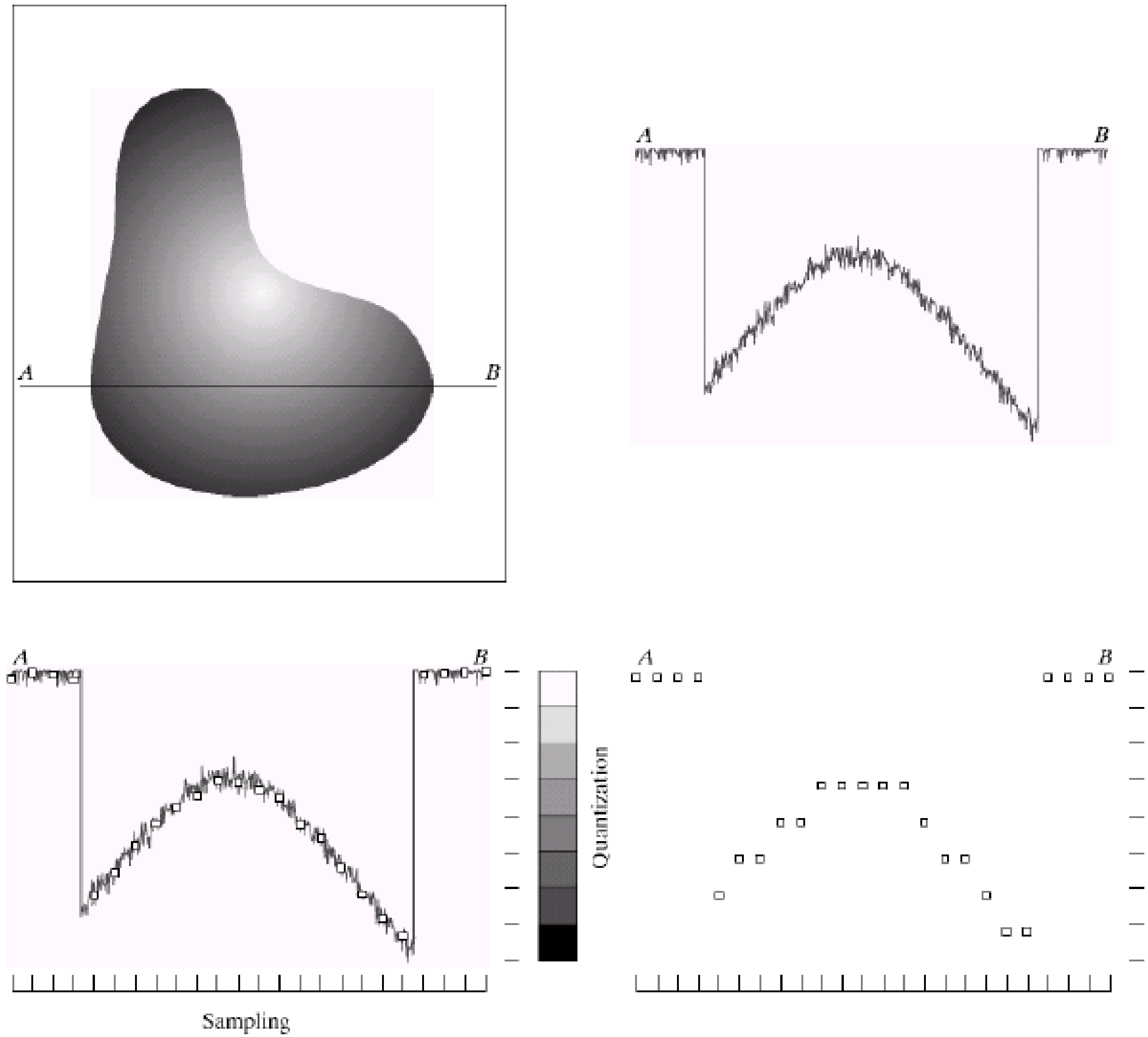
Image Sampling and Quantization

- Image sampling is the process of converting a continuous image (analog) into a discrete image (digital) by selecting specific points from the continuous image.
- Image quantization is the process of converting the continuous range of pixel values (intensities) into a limited set of discrete values.

Image Sampling and Quantization

- Image sampling refers to discretization of spatial coordinates (along x axis) whereas quantization refers to discretization of gray level values (amplitude (along y axis)).
- Given a continuous image, f(x,y), digitizing the coordinate values is called sampling and digitizing the amplitude (intensity) values is called quantization

Image Sampling and Quantization



Representing Digital Image

- An image may be defined as a two-dimensional function, f(x, y), where x and y are *spatial* (plane) coordinates, and the amplitude of f at any pair of coordinates (x, y) is called the *intensity* or *gray level* of the image at that point.
- We can represent M*N digital image as compact matrix as shown in fig below

$$f(x,y) = \begin{bmatrix} f(0,0) & f(0,1) & \cdots & f(0,N-1) \\ f(1,0) & f(1,1) & \cdots & f(1,N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1,0) & f(M-1,1) & \cdots & f(M-1,N-1) \end{bmatrix}.$$

Spatial and Gray-Level Resolution

- Spatial resolution is the smallest discernable change in an image.
- Gray-level resolution is the smallest discernable change in gray level.

Fig: A 1024 x 2014, 8 bit image subsampled down to size 32 x 32 pixels. The number of allowable kept at 256.

Spatial and Gray-Level Resolution

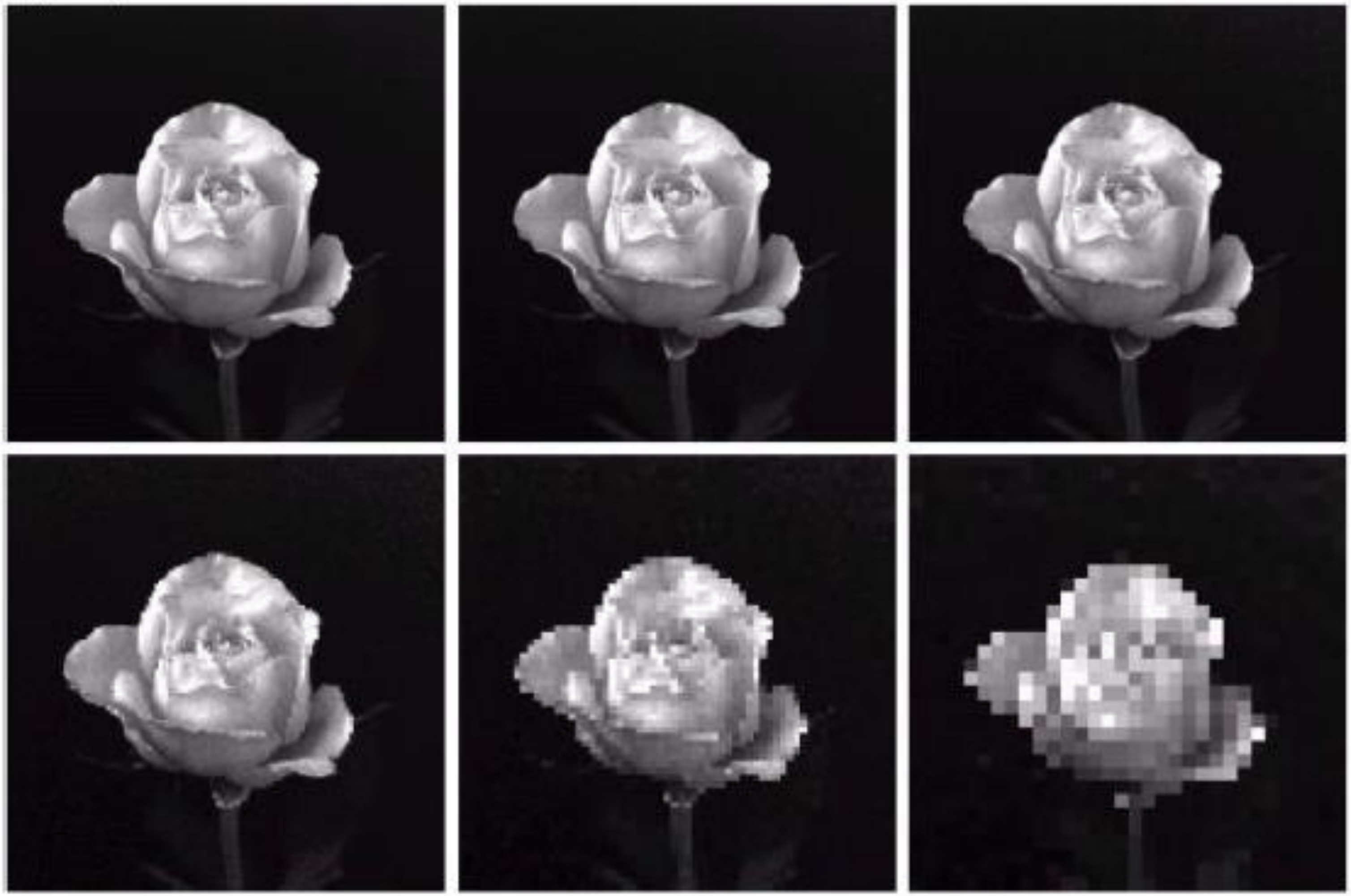


Fig: (a) 1024 x 1024, 8 bit image (b) 512 x 512 image resampled into 1024 x 1024 pixels by row and column duplication. (c) through (f) 256 x 256, 128 x 128, 64 x 64, and 32 x 32 images resampled into 1024 x 1024 pixels.

Spatial and Gray-Level Resolution

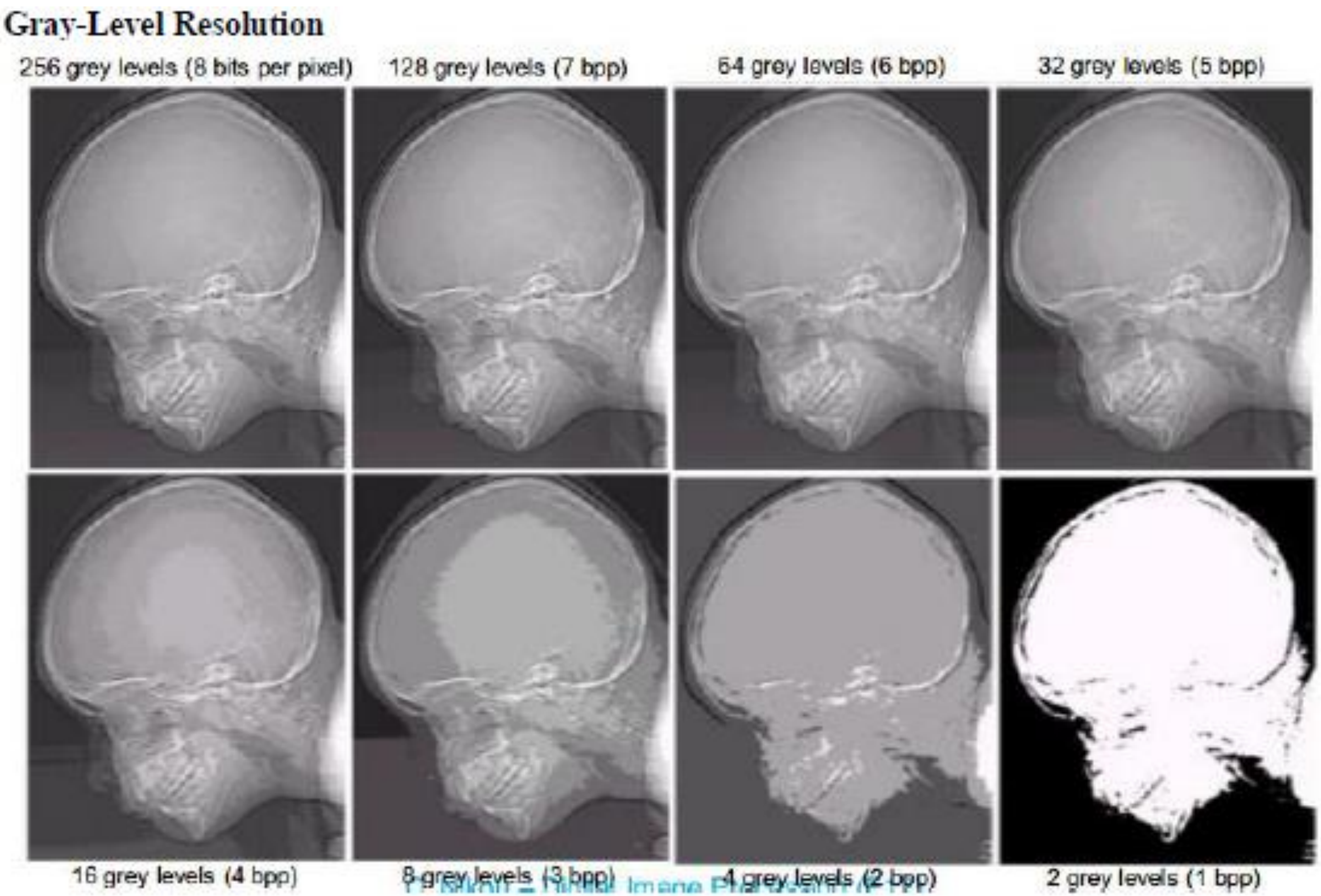


Fig: 452 x 374, 256 level image. (b)- (d) Image displayed in 128, 64, 32,16,8,4 and 2 grey level, while keeping spatial resolution constant.

Application of Image processing

- Healthcare & Medical Imaging: Revolutionizing Diagnostics
- Automotive & Autonomous Systems: The Eyes of Self-Driving Cars and Drones
- Security & Surveillance: Enhancing Safety with Facial Recognition and Anomaly Detection
- Retail & E-commerce: Transforming Shopping with Visual Search and AR Try-Ons
- Agriculture (AgriTech): Enabling Precision Farming through Crop and Soil Monitoring
- Industrial & Manufacturing: Automating Quality Control and Defect Detection
- Geospatial & Remote Sensing: Analyzing Environmental Changes and Urban Planning
- Entertainment & Social Media: Powering Filters, AR Effects, and Content Moderation